

Explainable Fake News Detection using BERT with SHAP and LIME

Abdullah (M24F0044DS009)

School of Computing Science

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology

Email: M24F0044DS009@paf-iast.edu.pk

Abstract—Fake news detection is a practical Natural Language Processing (NLP) problem where high predictive performance is not sufficient on its own; users also need to understand *why* a model predicts an article as fake or real. This report presents an end-to-end pipeline for fake news classification using a fine-tuned BERT model and two complementary explanation techniques: SHAP for token-level contribution analysis and LIME for local, instance-level explanations. On the held-out test set, the trained model achieved 0.91 accuracy. For the *Real News* class (positive label), the model achieved precision 0.703, recall 0.678, and F1-score 0.690.

Index Terms—Fake news detection, BERT, Transformers, Explainable AI, SHAP, LIME, NLP.

I. INTRODUCTION

Misinformation spreads quickly on online platforms and can influence public opinion, health behavior, and political stability. Automatic fake news detection aims to classify news content as fake or real using statistical patterns in text. Transformer-based language models such as BERT provide strong performance due to contextual representations, but their decisions are difficult to interpret. To make the system more transparent, this project integrates explanation methods that highlight influential words and phrases supporting each prediction.

II. DATASET AND PREPROCESSING

The project uses a preprocessed short-text fake news dataset stored as CSV files for training and testing. Each sample contains two columns: `text` and `label`, where label 0 corresponds to *Fake News* and label 1 corresponds to *Real News*. The dataset is imbalanced, as shown in Fig. 1.

A. Data Summary

Training data contains **25336** samples and the test set contains **6525** samples. A simple length analysis was conducted after preprocessing: mean text length is **86.54** tokens/words (depending on the preprocessing step), with median **81**, minimum **1**, and maximum **276**. These statistics guided the tokenization maximum length used for BERT.

III. MODEL: BERT FINE-TUNING

The classifier is built using `bert-base-uncased` with a sequence classification head for two classes. Given an input sequence $x = (w_1, \dots, w_n)$, BERT produces a contextual embedding for the special [CLS] token. A linear layer maps

TABLE I
DATASET SUMMARY AND TEXT LENGTH STATISTICS

Item	Value
Training set size	25,336
Test set size	6,525
Mean text length	86.54
Median text length	81
Min / Max text length	1 / 276
Labels	0: Fake, 1: Real

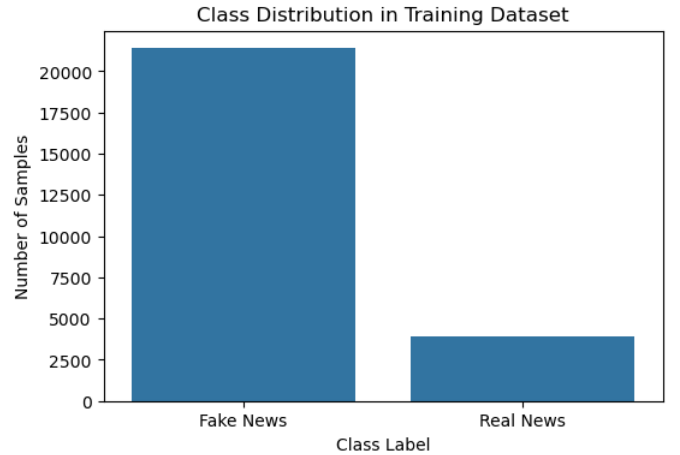


Fig. 1. Class distribution in the training dataset (imbalanced).

the [CLS] representation to logits $\mathbf{z} \in \mathbb{R}^2$, followed by softmax to obtain class probabilities. The model is trained by minimizing cross-entropy loss.

A. Tokenization

Texts are tokenized using the BERT tokenizer with truncation and padding, using maximum sequence length **128**.

B. Training Configuration

Training was performed using the HuggingFace Trainer API. Key hyperparameters are listed in Table II.

IV. EVALUATION AND RESULTS

Evaluation was performed on the held-out test set using standard classification metrics and a confusion matrix. Because the dataset is imbalanced, per-class precision/recall are reported along with overall accuracy.

TABLE II
FINE-TUNING CONFIGURATION (TRAINING ARGUMENTS)

Hyperparameter	Value
Model	bert-base-uncased
Max sequence length	128
Learning rate	2×10^{-5}
Epochs	3
Train batch size	16
Eval batch size	16
Weight decay	0.01
Evaluation / checkpointing	per epoch
Best model selection	enabled (load_best_model_at_end)

TABLE III
KEY TEST METRICS (REAL NEWS AS POSITIVE CLASS)

Accuracy	Precision	Recall	F1-score
0.91	0.703	0.678	0.690

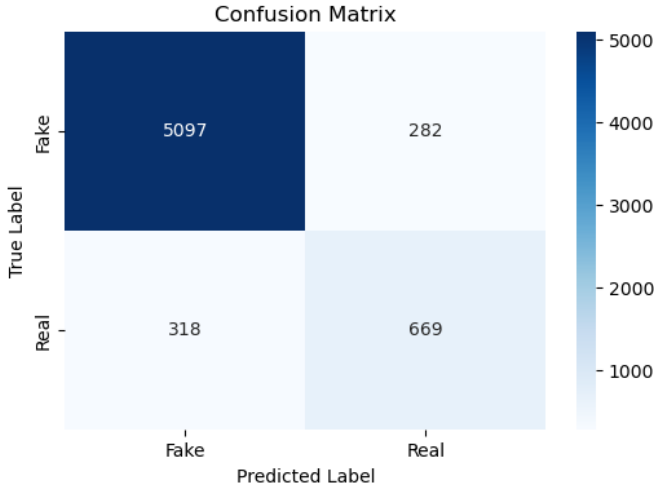


Fig. 2. Confusion matrix on the test set. Counts: TP/FP/FN derived from model predictions.

TABLE IV
CONFUSION MATRIX VALUES (TRUE LABEL $\times$ PREDICTED LABEL)

True \ Pred	Fake	Real
Fake	5097	282
Real	318	669

A. Overall Metrics

The model achieved overall accuracy of **0.91**. For the *Real News* class (label 1), precision is **0.703**, recall is **0.678**, and F1-score is **0.690**.

B. Confusion Matrix

Fig. 2 shows the confusion matrix. The matrix indicates strong performance on *Fake News* samples, while *Real News* is harder due to fewer examples and overlapping language patterns.

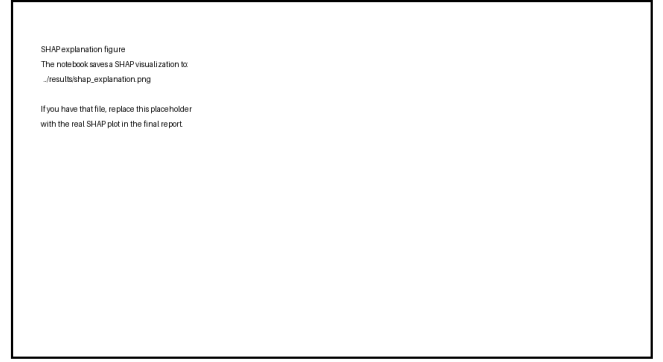


Fig. 3. SHAP-based explanation visualization (placeholder included for compilation). Replace with the saved project output shap_explanation.png.

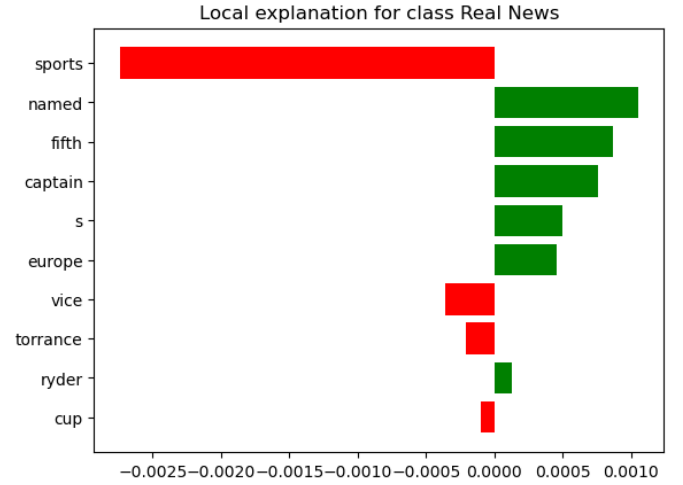


Fig. 4. LIME local explanation for a sample predicted as Real News.

V. EXPLAINABILITY: SHAP AND LIME

To increase trust and interpretability, this project applies SHAP and LIME to explain model predictions at the word/token level.

A. SHAP Explanations

SHAP assigns contribution values to tokens by approximating Shapley values, indicating how each token pushes the prediction toward *Fake* or *Real*. In practice, SHAP helps identify recurring influential patterns (e.g., sensational wording) and provides local explanations for individual samples. Fig. 3 is a placeholder image; in the project repository, the SHAP notebook saves the actual plot to `./results/shap_explanation.png`.

B. LIME Explanations

LIME explains a single prediction by perturbing the input text (removing or masking words), fitting a simple surrogate model locally, and reporting which tokens increase or decrease the probability for a target class. Fig. 4 shows an example local explanation for the *Real News* class.

